NIFTY intraday VRP research – highlights

An audited data foundation, an honest hypothesis rejection, and one frozen shadow candidate

Decision. Standalone 60–180 minute VRP timing is rejected after costs. One post-hoc p85 short-iron-fly candidate is retained for forward shadow only; current live allocation is Rs. 0.

AUDITED GOLD	SESSION COVERAGE	H4 EXECUTIONS	HISTORICAL NET
43.02m rows	**1,371 / 1,371**	**86 of 132**	**Rs. 108,983**

What we built

Layer	Delivered evidence
Acquisition	Dhan rolling options, NIFTY spot and India VIX with immutable request/response manifests.
Archive	Bronze provider bytes, typed silver Parquet and linked rejects, audited research gold.
Cleaning	Same-session backward joins, date-effective expiry/lot rules, anomaly gates and duplicate quarantine.
Volatility	Independent gold BSM plus parity-forward Black–76 research IV; provider IV is comparison-only.
Margin	Entry-time six-slot SPAN research schedule; historical file-arrival timing remains unproven.
Execution	Date-aware Groww charges, adverse liquidity/staleness slippage and corrected volume/OI quantity impact.

Research boundary

NIFTY single nearest-listed-expiry *proxy*; rolling ATM±10 source, structures inside ATM±3 at entry, defined-risk only, next-minute entry and fixed 60-minute exit. Strict fixed-contract path coverage supports ATM±3 through 180 minutes, but not unbiased multi-day inference. Historical bid/ask, queue position, intratrade MTM and proven SPAN arrival timestamps are unavailable.

What the hypotheses said

ID	Test and result	Decision
H1	Zero-cross short condor: 895 trades, Rs. 22.60 gross vs Rs. 271.14 cost; −Rs. 248.54 net/trade.	Reject
H2	70–95 percentile crossings up/down: all 12 one-lot variants negative net.	Reject
H3	Dynamics, structures, 60/120/180m, feature rescue and reversals: 0/48 structure-horizon cells positive net.	Reject
H4	Upper-85 up, gated short iron fly: selected one-lot mean Rs. 292 gross, Rs. 240 cost, Rs. 52 net.	Shadow

Matched intraday VRP

$$F_t = \mathrm{median}_K\{K + e^{rT}(C_t - P_t)\},$$

$$VRP_{t,60} = \widehat{\sigma}^2_{IV,t} - \frac{\sum_{i=1}^{60} r_i^2}{60/525600}.$$

The causal percentile compares only prior dates at the same minute of day; velocity and acceleration are changes in its trailing five-minute median. Signals never use future returns or later rolling labels.

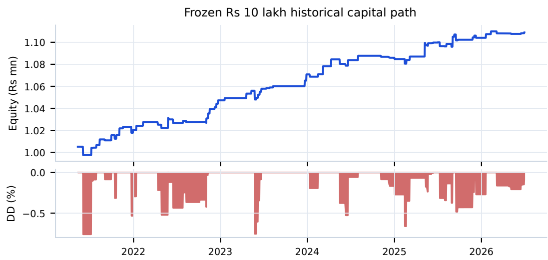
Why H4 remains conditional

The candidate is post-selected. Its combined 2024–26 score/P&L Spearman is 0.214, but the bootstrap 95% interval is $[-0.066, 0.484]$. Event weeks and post-20-Nov-2024 trades are weaker, and the 2024 slice has only 11 executions.

Rs. 10 lakh historical sizing snapshot

Metric	Result	Metric	Result
Gross / costs	Rs. 153,013 / Rs. 44,029	Net / return	Rs. 108,983 / 10.90%
Average / max lots	6.50 / 14	Margin use	31.93% avg / 34.97% max
Hit / profit factor	66.28% / 2.537	Avg win / loss	Rs. 3,156 / −Rs. 2,445
Max rupee DD	−Rs. 7,982	Trade CVaR 5%	−Rs. 6,142
Turnover / cost drag	Rs. 6.67m / 28.77%	CAGR / Sortino	2.04% / 2.75

The 35% SPAN and 4% cost-reserved max-loss caps produced a feasible profile, not a proven risk-return efficient frontier.



Next gate. Freeze every parameter and shadow one modeled lot. Require at least 100 new non-overlapping trades over 12 months, positive net in both six-month halves, score-rank bootstrap lower bound above zero, observed slippage within 1.5× model over 20 fills, forward drawdown below 1.5%, and no month above 40% of profit before discussing live capital.

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